

Institutions and Economic Development

Compare a simple transaction in a developed and a developing economy—the purchase of land by a farmer.

In a developed country, land ownership is protected by law, real estate transactions and land surveys are thoroughly recorded, and a dispute over a deed can be resolved in court. A registered title to land can be used as collateral to obtain a loan to finance improvements to the land. In a developing economy, however, this simple transaction can be fraught with many problems. There may be no system to track ownership titles and no surveys of a property's exact boundaries, so it can be difficult for a poor and often uneducated farmer to defend himself against encroachment by neighbors, who may destroy his crops or dig trenches within the borders of his newly purchased land. Local authorities may evict the farmer without adequate compensation in order to build roads or housing projects. This lack of legal protection tends to support rich and influential landowners and government officials. When a farmer cannot document ownership of his land, a large investment in seed, fertilizer, livestock, or equipment is a risky venture. The farmer will thus be less likely to contribute to the economic development of his country.

This simple example illustrates the importance of institutions that facilitate economic transactions and, thus, economic development. Although we have mentioned various institutions in previous chapters, we have not yet explored them in detail and what effect they can have on economic performance and development.

The economic concept of **institution** is defined as the constraints placed by law and *social norms* on human behavior. These constraints help reduce **transaction costs**, the costs associated to simple transactions such as buying or selling merchandise, borrowing money, or investing in a business. Every day, there are millions of transactions by consumers and producers within an economy. Just as tiny streams converge to form a great river, transaction costs contribute to a powerful effect on the aggregate performance of an economy and its growth rate. This chapter examines transaction costs in detail and how they are affected by institutions.

We begin our discussion of institutions by defining precisely what they are. We then discuss what institutions can do to reduce transaction costs. We compare institutional development in advanced economies and in developing economies, and get a better understanding for why a lack of more advanced institutions makes transaction costs higher in less-developed countries and creates serious obstacles to their economic development.

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What Are Institutions?

Following the example of Douglass North, Nobel Prize-winning economist, we define institutions as constraints on behavior imposed by societal rules: "Institutions include any form of constraint that human beings devise to shape human interaction."¹ This broad definition includes all constraints imposed on individuals and groups by a society's system of beliefs and values.

People who live in the advanced economies of the world often take institutions for granted. Even mainstream economics has long ignored them; it explains *how* the price mechanism helps balance supply and demand, but says very little about the institutions underlying efficiently functioning markets. A simple transaction such as a purchase of a commodity can be fraught with uncertainty without an adequate institutional environment. A commodity purchase requires the transfer of that commodity from the seller to the buyer and the transfer of payment from the buyer to the seller. If the two operations are not synchronized, there may be problems, depending on who moves first. If the buyer makes the payment first, the seller may never show up to deliver the goods, but if the seller transfers the goods first, the buyer may never bring the payment. Even if the operations in a transaction are synchronized, cheating is possible. The buyer may write a check that will bounce or the seller may hide important information from the buyer about the product sold.

The more sophisticated the transaction, the more opportunities there are for cheating and predatory behavior. Without adequate institutions, transactions are much more costly and the volume of transactions will then be much lower and less economic development will take place.

Formal Institutions

In some societies, **formal institutions** are codified in writing and are generally more precise than unwritten rules. Speed limits that specify how fast we can drive and electoral laws that incorporate the procedures for translating votes in an election into seats in a legislature are examples of formal institutions. Authorities can enforce formal rules with established statutes, which are also officially recorded. When talking about a country's formal institutions, it is useful to distinguish between *political* and *legal* institutions.

Political institutions. Political institutions are the set of rules that define the powers of the major political bodies and players (chief executive, legislators, etc.), and how they are selected. Political institutions determine the differences between political regimes, such as democracies and dictatorships. Democratic institutions make it possible to oust politicians who have misbehaved or change leaders when ideological preferences among the electorate have shifted, options that are not possible under a dictatorship. Usually, democracies incorporate separation of powers, whereas dictators hold all the power in

¹Douglass North, *Institutions, Institutional Change, and Economic Performance* (New York: Cambridge University Press, 1990): 3.

a government. Therefore, the rule of law is usually better enforced in democracies than in dictatorships. This is not always the case, however; there can be imperfect democracies as well as beneficial dictatorships. For example, in several Asian countries, such as Singapore, there are political regimes that are not democratic, but they still uphold the rule of law. We will study political institutions in more detail in Chapter 9.

Legal institutions. Non-political institutions codified by laws, legal institutions across the globe differ in many ways, and these differences can have important economic effects. Some legal institutions are strongly rooted in religion as is the case in certain Islamic countries where Sharia law, expressed in the Qur'an, is both religious law and moral code, and in Israel, where the Talmud is the foundation of many laws. Ancient cultures such as the Chinese or the Hindu civilizations have developed their own legal traditions based on customs that date back thousands of years.

Legal institutions not derived from religious traditions fall into two main categories: the common law and the civil-law systems. The **common-law system**, developed in England and other Anglo-Saxon countries (Canada, Australia, and New Zealand for example), is based on customary Saxon law, the law that existed in England before the Norman invasion of 1066. Legal arguments in the common-law system rely heavily on legal precedents, decisions made in past court cases. **Civil-law systems** rely mostly on recorded legal statutes and have developed from Roman civil law. Legal arguments in the civil-law system are made on the basis of the consistency or inconsistency of a decision with the principles laid down in statutory law.

These different legal systems are important to economic development because the legal systems in many developing countries were inherited from former colonizers. British colonies inherited the common-law system, while Spanish and French colonies inherited the civil-law system. In Chapter 10, we will examine legal institutions.

Informal Institutions

Informal institutions are those that are not legally codified; they are based on **social norms** (or **conventions**), accepted and expected standards of behavior. They are enforced in an informal way using peer pressure, feelings of guilt, and moral obligations. Even when a country is ruled by formal institutions, informal institutions still play a large role in terms of politeness and civility, hospitality, courtship, traditions, holiday celebrations, and rites of passage. Even in a country where formal laws exist, disputes between citizens are often solved first by referring to mutually understood and accepted social norms. When informal institutions cannot solve a problem, people turn to formal institutions.

Informal institutions are based on a country's **culture**, the set of beliefs and values passed down from generation to generation. Beliefs concern nature, interactions between humans and nature, and interactions among humans in a society. **Values**, the intrinsic worth a culture or society places on certain behaviors and beliefs, determine the set of social norms in a society. Religion is a

fundamental part of culture and encompasses most of the beliefs, values, and social norms in many societies. We will talk more about culture in Chapter 11.

Interactions between Institutions

There are complex interactions between formal and informal institutions. First, laws often evolve with culture and social norms. The death penalty, for example, was a popular method of punishment until many countries did away with it as a result of rising popular opposition.

Second, it is very difficult to impose formal laws or formal institutions that are not in accordance with existing social norms; conflict between formal institutions and social norms can create conflict in developing countries. For example, international borders established in Africa after decolonization did not correspond to existing ethnic subdivisions. Tribes were separated across formal borders, but (informal) tribal and ethnic loyalties took precedence over (formal) national loyalties. As a result, politics polarized along ethnic lines, resulting in violence. Legal systems inherited from former colonial powers conflicted with social norms associated with tribalism. Even though there were formal rules prohibiting nepotism, politicians favored their families and friends because this favoritism was seen as the fulfillment of long-established moral obligations to fellow tribe members. Similarly, in a country where social norms are based on religion, it is very difficult to impose laws that appear to violate religious obligations, prohibitions, or constraints. A case in point is the prohibition of the Islamic veil in schools and public places in certain countries, which is seen by practicing Muslims as violating their religious norms.

Third, formal institutions often strengthen the effects of informal institutions. The social norm of honesty is supported by a legal system punishing a dishonest trader and makes that norm easier to enforce throughout the society. Similarly, informal institutions often complement and reinforce formal institutions; legal codes may be incomplete and social norms can help to supplement them and make them more effective. In some legal systems, citizens are obligated by law to help someone who has had an accident or whose life is in jeopardy. Even when the law does not specify in detail when this legal obligation applies, social norms provide the moral obligation to help a person in danger.

Formal and informal institutions tend to be complementary and to have increased effectiveness in a society or culture when they are based on the same fundamental set of beliefs. If they are in conflict, formal institutions do not work as well as established social norms.

What Do Institutions Do?

As we discuss the economic role of institutions, we will examine five important problems that occur in basic economic transactions and show how institutions can help to solve these problems: informational problems, hold-up problems, commitment problems, cooperation problems, and coordination problems. The solutions to these problems have significant economic effects on the development of commerce, entrepreneurship, trade, innovation, and investment.

Efficient institutional solutions provide critical reductions in transaction costs and contribute to economic development.

Informational Problems

A simple transaction often involves informational problems, as one party is bound to have more information about a product's quality than the other party does, a situation that economists call **asymmetric information**. For example, food distributors may ship food products that were not stored under proper conditions, a used car salesman might sell cars with hidden mechanical problems, a software company may sell an operating system that has programming flaws, or a real estate agent may sell a house that has hidden structural defects. In most cases, the buyer may not have the information necessary to decide whether or not to purchase an item and/or it might be very costly or outright impossible to acquire all the relevant information prior to making a purchase. These informational problems are present in all economies, both developed and less developed. Lack of information about a product and a possible lack of trust in the seller may prevent the buyer from completing the transaction. Because of asymmetric information, many transactions, even legitimate ones, might not take place at all.

Asymmetric information is also detrimental to transactions that do not involve the exchange of goods. This is the case for credit contracts, where there is usually an informational asymmetry between the lender and the borrower. The lender does not know the **creditworthiness** of the borrower, his or her willingness or ability to fulfill a debt contract. Because of asymmetric information, a creditor may refrain from lending even to a very creditworthy individual. Entrepreneurs who have good ideas for the productive use of capital may not be able to realize those ideas because they might be unable to obtain loans to start their businesses. This is a very serious problem in developing economies.

Adverse selection. Informational problems may lead to **adverse selection**, a situation caused by informational asymmetries in which reputable lenders and sellers leave the market to disreputable players, resulting in a possible market collapse. George Akerlof received the Nobel Prize in Economics for his article "The Market for Lemons," in which he showed how informational asymmetries and adverse selection can cause the disappearance of markets or even severe market failure.² He described second-hand car markets in which the seller has information about the quality of the car but the buyer does not. Assume that there are good second-hand cars that cost the seller \$10,000 (purchase plus repairs) and are worth \$12,000 to consumers. Also assume that there are bad second-hand cars that cost the seller \$5,000 but are worth only \$4,000 to the buyer. If there are no informational asymmetries, good cars would sell and deliver a profit of \$2,000 to the seller and bad cars would not sell because they would deliver a loss of \$1,000. Assume now that there is informational asymmetry and that 50% of the cars are good and 50% are bad. A buyer now faces a 50% chance of buying a car that is worth \$12,000 and a 50% chance of buying a

²George A. Akerlof, "The Market for 'Lemons': Quality Uncertainty and the Market Mechanism," *The Quarterly Journal of Economics* 84, no. 3 (1970): 488–500.

car that is worth \$4,000. A buyer who is **risk neutral**, who values upside and downside payoffs equally and weighs them according to their probability, would not be ready to pay more than the **expected value** $((0.5 \times 12,000) + (0.5 \times 4,000) = \$8,000)$ for a car. In other words, a risk-neutral agent looks at the **expected payoff** of a decision: the downside payoff (\$4,000) times its probability (.5) plus the upside payoff (\$12,000) times its probability (.5). A **risk-averse** buyer, who puts less weight on the upside payoff than on the downside payoff for given probabilities, would pay even less or would possibly not even be ready to pay more than \$4,000 to buy the car.

In this situation, the seller of a bad car would make a profit of \$8,000, but the seller of a good car would suffer a loss. Compared to a situation of perfect information, asymmetric information would tend to drive out the reputable sellers from the market and attract the disreputable sellers. If the honest sellers withdraw because they cannot make a profit, buyers will soon realize that only bad cars are available for which they are only willing to pay \$4,000, and the market will disappear.

Adverse selection also occurs in the credit market. Suppose that there are two types of borrowers, good and bad. The good borrower is creditworthy and needs financing for a project that will earn a 10% return. The bad borrower does not intend to repay the loan at all. The bank must pay depositors an interest rate of 6%, otherwise they will withdraw their deposits and put their money elsewhere. Note that the return on the project for the good borrower (10%) is higher than the market return for depositors (6%). Assume that among the population of potential borrowers there is a 50% chance that a potential borrower is good and a 50% chance that he or she is bad. Without asymmetric information, the bank would not lend to the bad types and would lend only to the good types, making a profit because it could charge the borrowers up to 10% interest. Under asymmetric information, however, the bank cannot discriminate between good and bad borrowers and would have to charge the same interest rate to all borrowers. In order to pay back its depositors, the bank must charge an interest rate of at least 12% (6% divided by 50%, the probability that a loan is paid back). This interest rate will be too high for the good borrowers, who will decide not to apply for a loan, but the bad borrowers would not be discouraged by the high interest rate, because they do not intend to pay back the loan. Good borrowers are thus driven out of the market by the excessively high interest rate.

Note that because of adverse selection, the price of credit, the interest rate, cannot serve as an efficient mechanism to adjust supply and demand in credit markets. If there is an excess demand for credit and if banks decide to adjust the interest rate upward to reduce that demand and attract more depositors, the effect is likely to be a decline in the quality of the credit pool, as some good borrowers whose projects have a somewhat lower profitability become discouraged by the higher interest rate while bad borrowers remain undeterred. The bank might reduce its profits by increasing interest rates because of the deterioration of the quality of the credit pool. For that reason, banks prefer not to increase the interest rate to market clearing rates and instead decide to ration credit (**credit rationing**). This means that only some applicants receive a loan and many other applicants are turned down. Obviously, banks try to gather as much information as they can on potential borrowers and use this information

to decide to whom they allocate credit. While this information search helps to alleviate informational asymmetries, it is usually only a partial solution to the problem: many creditworthy borrowers are still denied loans while untrustworthy borrowers are approved for credit.

Moral hazard. Another problem that arises from asymmetric information is **moral hazard**, the possibility of unobservable actions one of the contracting parties can take that will hurt the interest of the other contracting party. A typical example of moral hazard arises in the relationship between a landlord and a wage laboring farmer. If the landlord pays the farmer an hourly wage to work the land, the farmer may not work as hard as he would if the land were his own because the landlord cannot observe how hard he works and he receives the same wage whether he works hard or not.

The difference between moral hazard and adverse selection is not always clear. Moral hazard involves asymmetric information as well as unobservable actions. Adverse selection involves only asymmetric information. Take the example of the automobile insurance market. Assume a situation in which there are good and bad drivers, but all drivers do their best to avoid accidents. This is a case of adverse selection. Drivers differ only in their abilities and insurance companies cannot, without prior investigation, distinguish between good and bad drivers when offering an insurance contract. Now assume a situation in which all drivers are equally good but they may decide to drive less carefully because they know they are insured. This is a case of moral hazard in which the action taken is how carefully to drive.

In the case of a credit relationship, the difference between adverse selection and moral hazard is blurred. Adverse selection occurs if the quality of the loan is independent of actions taken by the borrower, as is the case if the borrower is intrinsically dishonest or incompetent. On the other hand, there is a moral hazard problem when the quality of the loan depends on the actions of the borrower. The borrower can deliberately choose a riskier project or fail to take the necessary actions to make the project profitable. In reality, a combination of both moral hazard and adverse selection exists, but it is important to keep in mind the conceptual difference between them.

Institutional solutions to informational problems. The obligation by one or more parties to a transaction to disclose information relevant to the transaction is referred to as **disclosure rules**, legal solutions to informational problems such as in the case of the sale of a house or a company that is listed publicly. The seller of a house must, for example, disclose information on past repairs or damages to the house. Executives of a company listed for sale must reveal information about their business dealings and about their governance. In many professions, there is **regulation of access or entry**, rules that govern entry into and the practice of a profession, and require a candidate to earn a qualified diploma or pass an exam as a means to screen out unqualified people. This is the case for doctors; without rules for entering the medical profession, consumers would have no formal way to distinguish between qualified and unqualified doctors. As a result, most countries strictly regulate the qualifications of medical professionals. This issue seems obvious, but in India, for example, many doctors (sometimes a majority of the practicing physicians in certain

areas) do not have a medical degree. These “doctors” are often completely incompetent: they are unable to make correct diagnoses and tend to prescribe unnecessary medicines.³

Other solutions to informational problems are the result of actions taken by firms that are sometimes backed by legal institutions. An obvious example is the association of a **warranty** (the obligation by a seller to guarantee the quality of a product) with a sales contract. Warranties take different forms, such as a guarantee to repair goods for free within a specified amount of time after the purchase of a good or after the replacement of a deficient component of that good. **Return policies** allow the buyer to bring purchased goods back to the seller within a specified time after their purchase. Warranties and return policies make it possible for the seller to give credible information about the product. If consumers can learn quickly about the quality or usefulness of a product, a return policy signals to a buyer that the seller is confident in the quality of the product. If it requires long-term use of a product to determine its quality, a warranty is a better way to signal quality to a consumer by shifting any repair risk onto the seller.

Sellers also use brand names to indicate quality. **Signaling** is an economic term that means taking certain actions to provide other parties in a transaction with credible information. In the case of a sales contract, it means giving credible information to consumers about the quality of a product. Consumers consider a well-known brand name with a reputation for good quality to be reliable, and they will trust products from that brand. The company that has established a brand name has an interest in keeping its reputation because it is the basis for consumer trust in its products.

Entering a franchise contract allows an entrepreneur to benefit from the reputation of a brand, which makes it possible to attract customers more easily. Franchising allows businesses to expand very easily, but firms need to monitor their franchisees, as loss of brand reputation can be quite damaging. In China, the developing world's largest economy, there has been an expansion of franchised fast-food chains in recent years with KFC and McDonalds from the United States, Ajisen Ramen or Saizeriya from Japan, or even Chinese fast-food chains such as Da Niang Dumpling and Ji Xiang Wanton. According to the marketing research firm A. C. Nielsen, food hygiene security is cited as the top reason for the popularity of fast-food chains in China, as opposed to restaurants. This concern is understandable; cases of food poisoning in Chinese restaurants have increased in recent years. For example, the Xinhua News Agency reported that in 2007, 258 people died from food poisoning in China, an increase of 32% over 2006. Some fast-food chains in China have failed precisely because they were not able to build a reputation for safe food. This was the case for Rong Hua Chicken, which tried to compete directly with KFC. The chain was started in 1994, but had closed by 2000 because it was unable to enforce strict rules on the quality of raw and processed foods, leading to safety concerns.

Chain stores also rely on the logic of brand recognition. We know familiar names such as CVS, Best Buy, or Walmart in the United States, but there are

³See, for example, Jishnu Das and Jeffrey Hammer, “Which Doctor? Combining Vignettes and Item Response to Measure Clinical Competence,” *The Journal of Development Economics* 78, no. 2 (2005): 348–383.

also famous chain stores in developing countries—Marayana in Indonesia or Sanborns in Mexico. Brand-name chain stores sell hundreds of different types of products and unreliable merchandise could hurt their reputations. Chain stores are not only sales intermediaries between brand-name firms and consumers; they are also **informational intermediaries** because they gather information about the products they sell before offering them in their stores.

Rating agencies also specialize in gathering information that is relevant for consumers or companies. Credit reporting companies, for example, gather information on borrowers and rate their creditworthiness. However, even in advanced market economies this reporting system is far from perfect. Larger credit rating agencies such as Moody's, Standard & Poor's, and Fitch, which rate the creditworthiness of large firms and securities, were accused of serious inaccuracies in the years preceding the global financial crisis of 2008–2009. They were heavily criticized for giving good ratings to mortgage-backed securities based on subprime mortgages and for severely understating the risks associated with those securities. This huge failure in the financial sector has highlighted the need for more stringent regulation of credit-rating companies.

Another example of a rating system is eBay's transaction feedback. When eBay started to expand its online sales in the late 1990s, users reported many cases of fraud. Because of the risk consumers take when purchasing an item online, eBay devised a rating system in which buyers and sellers review a transaction after its completion. These ratings help members evaluate buyers' and sellers' reputations for good faith and honesty.

It is not a coincidence that many of the examples of contractual solutions to informational problems are from advanced economies rather than developing economies. These solutions only work well with the support of efficiently functioning formal institutions. In a country where the legal system is inadequate, where law enforcement is inconsistent, and where the court system is inefficient or corrupt, warranties give less protection than they do in a country where there are strong legal institutions. A dishonest seller can use a warranty to attract customers but then break his promise to provide a quality product. If disgruntled customers cannot use the court system effectively to protect their rights, then warranties are meaningless.

Informal solutions to informational problems. When formal institutions are deficient, which is most often the case in developing economies, informal institutions emerge to deal with informational asymmetries. If courts are not effective, repeated interactions and reputations are the main instruments buyers and sellers can rely on to overcome informational asymmetries.

When informational asymmetries exist with no formal institutions to help overcome them, market competition becomes less effective even when there are many buyers and sellers. As a result, buyers and sellers will enter into long-term relationships with a network of trusted personal contacts. This is called **relational contracting**. Buyers will have fewer incentives to search for sellers offering lower prices or even to do business with them because the buyers will not know whether a product is of lower quality than what they can obtain from their usual suppliers. Similarly, sellers will not have the incentive to search for buyers who are willing to pay a higher price because they do not know whether alternative buyers will uphold their promise to pay. For example, in Madagascar's

agricultural markets, the largest trading firms always do business with the same suppliers and customers.⁴ Having a large number of these time-tested relationships is seen by the traders as the biggest determinant of success in business. These business contacts are not only used to exchange information, but also to share risks and to borrow or lend money when necessary. In the 1990s, during the turbulent period of transition in Russia's market economy, legal institutions were very weak; as a result, firms developed repeated business relationships with a fixed set of suppliers and clients and were reluctant to enter into new business relationships.⁵

In a situation of relational contracting, breach of trust can be fatal to the relationship. If a partner deviates from honest behavior, his or her counterpart could end the relationship immediately, forcing the now-unreliable partner to face the uncertainty of the market environment. Because the prospect of finding another reliable trading partner in the market is fraught with informational asymmetries, the threat of terminating the relationship can serve as a deterrent to opportunistic behavior.

Less competition for new partners, and buyers and sellers stuck in unprofitable relationships creates an environment that will hinder market growth. The expansion of markets is stronger when formal institutions work to enforce contractual solutions to informational asymmetries.

Reputation also plays a role when formal institutions are deficient. Economic agents who build a reputation of honesty, reliability, and efficiency will benefit because their reputations will make them desirable business partners. Reputable buyers and sellers have fewer incentives to indulge in opportunistic behavior due to the significant investments of time and effort necessary to acquire their good reputations.

However, in order for reputations to be effective when formal institutions are deficient, there has to be an adequate and efficient circulation of information. Without this movement, information about reputations does not move through the marketplace. In the modern world, communicating this information is easy because it is available instantaneously on the Internet, whether through e-mail, blogs, or social media. Unfortunately, poor countries are at a disadvantage because information technology is not well developed in them. In many of these countries, the primary means of disseminating information is still by word of mouth. Because of inefficient communication between buyers and sellers, the incentives to build a reputation are weaker and the incentives to behave opportunistically are stronger. Developing countries are more likely to have "fly-by-night" traders and itinerant swindlers. For instance, in Nigeria there are many small retail outlets or vendor booths that rotate through a region's local markets, selling shoddy electronic goods. Consumers often spend a large part of their household budgets on faulty electronic products that break down after a couple of weeks. These products are sold without warranties and consumers have no hope of receiving a refund. There are countless

⁴Marcel Fafchamps and Bart Minten, "Relationships and Traders in Madagascar," *The Journal of Development Studies* 35, no. 6 (1999): 1–35.

⁵Kathryn Hendley, Peter Murrell, and Randi Ryterman, "Law, Relationships, and Private Enforcement: Transactional Strategies of Russian Enterprises," *Europe-Asia Studies* 52, no. 4 (2000): 627–656.

similar examples of opportunistic sellers on markets in developing countries who contribute to consumers' lack of trust and slow the development of reliable markets, putting poor countries at a double disadvantage. Not only are developing countries more likely to have weaker formal institutions, but the informal institution of reputation is less effective without the support of formal institutions and without good circulation of information.

The Hold-Up Problem

A widespread problem in business contracts, the **hold-up problem**, refers to a situation in which a business partnership requires an investment from one of the parties that is specific to the transaction in question. Once the investment has been made, however, the other partner "holds up" the party that made the investment in order to renegotiate the deal.

For example, suppose a truck manufacturer in India agrees to subcontract the production of axles to another party. An investment of \$10,000 is needed to produce the axles and it costs \$20 to produce each axle, including the cost of capital. The truck manufacturer contracts an order of 1,000 axles and promises a price of \$30 per axle. At that price, the transaction, based on the order of 1,000 axles, makes it possible for the axle subcontractor to break even, since he will be able to recover both the investment cost and the production costs. Indeed, the cost will be $\$10,000 + (\$20 \times 1,000) = \$30,000$ and sales will be \$30,000.

Once the axle company has made the investment in the machines to produce axles, however, the truck manufacturer tells the subcontractor that he is now ready to pay only \$25 for an axle. What does the subcontractor do? He has the choice of either producing the axles at the lower price of \$25 or not producing at all. If he does not produce, he loses \$10,000, but if he goes ahead with production, he will only lose \$5,000 (his operating profit will be $\$25 \times 1,000 - (\$20 \times 1,000) - \$10,000 = -\$5,000$); as a result, the subcontractor will prefer to produce the axles. Because the investment cost is already sunk (**sunk cost**), the only thing that matters in deciding whether or not to produce the axles is the difference between sales ($\$25 \times 1,000$) and operating costs ($\$20 \times 1,000$), which is \$5,000.⁶

Because the subcontractor will decide to produce, even though he will incur a loss, the truck manufacturer can take advantage of the subcontractor's situation when proposing the lower price. The truck manufacturer is holding up the subcontractor, taking advantage of the subcontractor's weaker bargaining position after he has made the investment. If the subcontractor had known the truck manufacturer was going to behave in this manner, then he would not have invested in the first place. A profitable investment opportunity can thus be lost because of the hold-up problem. In general, hold-up problems cause underinvestment, which is economically damaging in particular in developing countries.

Hold-ups are not confined to specific industries. Suppose the representative of a food company convinced a farmer to change crops and produce cocoa,

⁶Remember the notion of sunk cost. Once a cost has been incurred and cannot be recovered, it should not count in economic calculations of future choices. We apply this principle here.

promising a high price once the crop was mature. The farmer changes crops, and for the first year, harvests cacao beans from which cocoa is produced. If the food company comes back and offers a lower price for the crop, the farmer is still forced to sell to the food company or face the loss of an entire year's income.

Another example occurs in the rental market. If a farmer who is leasing land from a landlord has made significant efforts to improve the productivity of the land by building irrigation systems using his own money and effort, he may be evicted by the landlord, who can charge a higher price to other farmers because of the improvements, or can threaten to evict the farmer unless he agrees to an increase in the rental price.

The hold-up problem is more acute the more **relationship specific** the investment is. In the axle example, the investment to produce axles only has value when the subcontractor sells the axles to the specific truck manufacturing company. When an investment only has value in the context of a specific partnership with another firm, this relationship specificity is also called **asset specificity**. If the axle producer can sell the same axles to another truck company, the hold-up problem is solved because there are alternative uses for the investment. If the truck manufacturer knows this, he may refrain from trying to lower the price after the investment is made. Even if the axles are specific to a particular contract, the hold-up problem might still be mitigated if the prior investment can be restructured at a relatively low cost for alternative uses. Suppose that the subcontractor producing the axles can restructure his facility to produce metal parts for the construction industry; he might opt to do that instead of submitting to the demands of the truck manufacturer to lower the price of the axles.

Note that the hold-up problem is not related to informational asymmetries. The axle producer knows that the truck manufacturer will have an interest in holding it up. The hold-up problem is related to asset-specificity, the sunk cost of investment and opportunistic behavior.

Institutional solutions to the hold-up problem. Binding and legally enforceable contracts are the primary solution to the hold-up problem. If the buyer is contractually obligated to buy products from the subcontractor at a specified price, the problem should disappear. Countries with strong legal institutions have an advantage over countries with weak institutions in terms of the hold-up problem.

However, legal institutions do not always provide full protection against hold-ups. Because contracts are often incomplete and lack specific details, it is often possible for a buyer to invoke a contingency that was not built into the initial contract and lay the blame on the seller. One tactic is to claim that certain characteristics of a product that were not actually specified in the contract are not as initially agreed upon. Many of these claims are often not very **credible**, but it can be costly to mount a defense against them in court. The truck manufacturer can threaten the axle subcontractor with legal proceedings unless the axles' price is renegotiated downward, and the subcontractor may not have the resources to defend himself. Between the costs of a lower axle price and the cost of a lawsuit, the subcontractor may choose the lower axle price. The less detailed the contract, the more possibilities there are for hold-up.

Informal solutions to the hold-up problem. If legal protections against the hold-up problem are insufficient, then a widespread solution is **vertical integration**, a situation in which two firms working together as supplier and client merge into a single firm. In the hold-up example above, vertical integration means that axle and truck production would be part of the same firm. The truck company would produce the axles it needs because the axle subcontractor does not trust the truck manufacturer enough to be ready to make an investment in a joint business deal. This solution might be inefficient and the cost of producing the axles inside the truck firm might be higher than using a subcontractor.

Because there are fewer legal protections in developing countries, firms should have more incentives to engage in vertical integration. Unfortunately, the obstacles to achieving vertical integration are also higher because it requires a high level of capital to operate properly. In a poor country, this level of capital is more difficult to achieve due to less pre-existing wealth and underdeveloped capital markets.

Most multinational corporations that operate in developing countries are vertically integrated and control the complete manufacturing process from raw materials to the final product. They do not face the severe capital constraints that confront domestic firms in developing countries because they are able to raise funds on international capital markets. Operating as vertically integrated units in developing countries is therefore less of a problem for multinational corporations than for domestic firms.

Social norms might also work to mitigate the hold-up problem by reducing the amount of opportunistic behavior, but it is debatable by how much. Economists are skeptical of the effectiveness of social norms to eliminate this behavior because they tend to think that economic agents pursue their own self-interest. Social norms work well only to the extent that there are credible punishment mechanisms for those who deviate from them.

The Commitment Problem

In all types of behavior, when one party reneges on prior commitments made to another party in the course of joint transactions, it is referred to as the **commitment problem**. This widespread problem occurs in all kinds of transactions but is potentially present in all **sequential transactions** that do not take place at the same time but rather in a certain order. Take the simple transaction of purchasing a bicycle. The transaction is simultaneous if the buyer delivers the money at the same time that the seller transfers ownership of the bike. The same transaction is sequential if ownership precedes the money transfer or vice-versa. In any sequential transaction, a commitment problem can occur; if the buyer is to pay after delivery of the bike, she may take it but fail to pay for it. Similarly, if delivery occurs after payment, the seller may renege on his promise to deliver the bicycle after he receives the buyer's money.

The commitment problem is very general and does not require any investment or sunk cost. Commitment problems arise because the optimal choice for one of the contracting parties at one point in time is no longer optimal at a later stage. Economists call this difference in the optimality of decisions at different periods in time **time inconsistency**. In our bicycle example, it is optimal to

promise to pay for the bicycle before its delivery, but it may no longer be optimal to pay once the delivery has occurred, especially if the buyer can renege with no consequences.

Consequences of the lack of commitment. The inability to commit hurts both parties in a transaction and is, in many cases, an impediment to transactions. There are also situations in which one party's inability to commit weakens its bargaining power and renders its threats not credible to the other party. For example, a government could commit to a non-negotiation policy with terrorists or pirates who take hostages, but might be unable to resist the pressure to negotiate once such a hostage situation occurs. This wavering of resolve weakens the government's bargaining power and invites further hostage-taking actions.

Credibility, a party's ability to uphold the commitments it makes, plays an important role in monetary and fiscal policy. A government might promise not to run fiscal deficits, but before an election, it could abandon that commitment by raising expenditures to garner votes. Lack of credibility in this case means that agents will expect fiscal deficits and behave accordingly. In particular, a central bank may want to announce a tight monetary policy, but may later engage in unexpected monetary expansion to stimulate the economy. The central bank's lack of credibility will lead agents to expect inflation to be higher than the targets announced by the central bank. A monopoly may be unable to commit to maintain current prices once it has sold to the customers with the highest willingness to pay, as has often been the case with new types of computers, DVD players, and with flat-screen TVs. Initially, these televisions were quite expensive and many consumers preferred to wait for the price to fall, secure in their knowledge that the manufacturers would drop their prices as soon as demand declined, regardless of announcements made to the contrary. In this case, if the monopoly could make credible commitments not to lower prices, consumers would not wait for the price to fall before making a purchase.

Institutional solutions to the commitment problem. When formal institutions are reliable, contract enforcement plays a crucial role in solving commitment problems. The parties draw up a contract that makes commitment legally binding and possibly provides specific penalties for breach of contract. If one party reneges on a commitment, the other can go to court to punish the guilty party. This "third-party enforcement" serves as a powerful instrument to solve commitment problems. Legally enforceable contracts thus help to "tie one's hands," which is usually beneficial for both parties.

Solving the commitment problem by legally binding contracts does not necessarily imply rigidity. A contract may specify actions that are no longer optimal for both parties. For example, two companies may have a delivery contract for a certain good over a certain period of time. If the producer comes up with a new product that is both less costly to produce and gives more utility to the buyer, the two parties can agree to void the existing contract or to draw up a new agreement. However, the existing contract still protects each party to the contract. If the two agree to void a contract, it must indeed be the case that both are made better off, and no one suffers from the decision.

The weakness of formal institutions in developing economies tends to make the commitment problem more prevalent than in developed economies.

Informal solutions to the commitment problem. In traditional societies, witnesses were usually present when promises were made between two parties in a trade agreement or other type of contract. Witnesses to a marriage commitment are an important institution in many societies throughout the world; today's "best man" and "maid (or matron) of honor" are participating in an ancient tradition of witnessing a binding contract.

Parties to agreements have often devised ingenious informal mechanisms to create credible commitments. This is the case in all spheres of life: trade, crime, and arms sales, as well as societal commitments such as marriage. For example, army generals have often signaled their commitment to fight on, regardless of the consequences, by burning bridges behind their troops so that the soldiers had nowhere to retreat. These actions have rarely failed to impress the enemy. In the business world, a company might undertake a very large investment that can only be recouped if the product becomes the "next great thing" with booming sales. The company undertakes the investment to show its commitment to challenge other market competitors. On a darker note, in Japan many businesses hire the *yakuza*, organized crime syndicates, as debt collection agencies. Yakuza agents who threaten a recalcitrant debtor will go so far as to cut off their own finger in front of the debtor to emphasize their commitment to recovering the debt.

Because formal contract enforcement is so important in solving commitment problems, developing countries with deficient formal law enforcement must often use informal enforcement agencies.

As is the case with hold-up problems, informal institutions are much less effective than formal ones when it comes to solving the commitment problem.

The Russian Mob in a Market Economy

Informal enforcement methods were widespread in Russia during the 1990s, when the country started its transition to a market economy, and they often involved criminal organizations or firms that had close links to organized crime. These connections were efficient at using the threat of violence to enforce commitments made by private agents in a variety of contractual situations. Obviously, there was a downside to the use of criminal organizations as enforcement agencies: their methods turned quickly into

abuse as contracting parties increasingly turned to the threat of violence as a means to renegotiate agreements. The criminal agencies engaged in racketeering by extorting money or favors from the firms that had hired them in the first place. Competition among criminal agencies led to a deepening spiral of violence and terror. When formal institutions in developing nations cannot enforce both criminal and civil laws, businesses may be forced to use criminal agencies to protect themselves and their business interests.

The Cooperation Problem

In a situation where individuals follow their self-interest in making their choices and where the outcomes of these choices are less than optimal for all parties, a **cooperation problem** arises. This means that if individuals cooperated to find a solution that would be better for all parties involved, they could achieve a better payoff.

The prisoner's dilemma. Economists represent the cooperation problem with the prisoner's dilemma, an important example of **game theory**, which studies decisions of agents with payoffs that depend on the actions of other agents.

In the **prisoner's dilemma**, two suspects have been arrested by the police and are interrogated in the attempt to extract confessions to a joint crime. Each prisoner is in a separate cell and cannot communicate with the other. As Table 7.1 shows, if neither confesses, they will both serve a maximum of 6 months in jail. If they both confess, they will each serve 2 years. However, if one confesses and the other does not, the one who confesses will be set free immediately while the other will spend 10 years in jail. What will happen? Game theory tells us that the **Nash equilibrium**, a situation in which each agent has chosen an action that is optimal given the action of the other, is the scenario in which both confess. A Nash equilibrium also occurs when one player does not want to deviate from his or her actions given the actions of the other.

Let us examine the strategies presented by the prisoner's dilemma. Suppose B confesses. It is then in A's best interest to confess also. If A does not confess, he will get 10 years, and if he does confess he will get 2 years. If B decides not to confess, it is again in A's best interest to confess. Indeed, if A does not confess, he will get 6 months, but if he does confess, he walks free. We see that whether B decides to confess or not, it is always in A's best interest to confess. This is called a **dominant strategy**, a situation in which the same action is optimal and independent of what the other player chooses. It is a **dominant strategy for Prisoner A to confess**. Since Prisoner B's reasoning is the same as Prisoner A's, Prisoner B will also decide to confess. Therefore, they will both serve 2 years.

They would both be better off if they had decided not to confess. In that case, they would each only get 6 months. However, this is not a Nash equilibrium. Suppose that they each decided not to confess. In that case, each would gain individually by changing his mind and confessing in order to walk free.

TABLE 7.1 The Prisoner's Dilemma

	B stays silent.	B confesses.
A stays silent.	Both serve 6 months.	Prisoner A serves 10 years. Prisoner B goes free.
A confesses.	Prisoner A goes free. Prisoner B serves 10 years.	Both serve 2 years.

Prisoners A and B decide individually whether to confess to a crime. It would be in their joint interest for both to stay silent, but if they follow their self-interest, they will both choose to confess ... and end up staying in jail longer.

Note that the commitment problem comes into play here. If both could precommit never to confess and were able to enforce such commitment, they would both be better off. However, if there is no mechanism to implement such precommitment, then it is in their self-interest to confess and the outcome is suboptimal. We can understand all cooperation problems in terms of the logical structure of the prisoner's dilemma.

The inefficiency of pure self-interest maximization. The prisoner's dilemma is often called a cooperation problem because it is a typical representation of a scenario in which each agent follows only his or her self-interest and where the outcome is suboptimal for everyone. If cooperation could be enforced (in this case, a precommitment not to confess), then it would be better for both parties. The prisoner's dilemma is often invoked as a symbol of the inefficiency of a pure laissez-faire system, in which individuals only follow their self-interest. This view is usually contrasted with traditional visions of laissez-faire, where following self-interest is best for society as a whole.

There are many relevant examples of the cooperation problem, such as protecting the environment. It is costly for any single country to take measures to prevent climate change; reducing carbon dioxide emissions means introducing new car models that run on low-carbon or carbon-free fuels, investing in public transport, and developing energy-saving technologies. If all countries took these measures simultaneously, then they could achieve tangible results. However, countries also face the prisoner's dilemma: if all other countries take measures to preserve the environment, then it is possible for a single country to benefit from the investment of the others without incurring the costs.⁷ If all other countries decide not to invest in averting climate change, then it is not in the interest of a single country to incur the necessary costs because these investments will not yield benefits if other countries fail to reduce their carbon emissions. It is thus a dominant strategy not to invest in reducing carbon dioxide emissions. This lack of cooperation can be a tragedy in the long run. As most environmental issues reflect the prisoner's dilemma, institutional solutions are needed to solve them.

One such solution came in the form of the **Kyoto Protocol**, a modest first step toward international cooperation to prevent climate change. The Kyoto Protocol required 36 developed countries to reduce their greenhouse gas emissions by 5% relative to their 1990 levels. This was considered by climatologists to be too modest a target. However, even that first step was far more than some countries were willing to take, and led by the Bush administration, the United States refused to sign the agreement. In December 2009, the United Nations' **Copenhagen Accord** failed to reach an agreement with binding commitments by participating countries to reduce their carbon emissions. This failure illustrates the cooperation problem and the difficulties of finding solutions to complex problems when the parties involved follow their own self-interests.

⁷This is not true for a very large country. The United States alone is responsible for about a third of carbon dioxide emissions in the world, so it would not benefit that much from free-riding on the environmental efforts of others since a U.S. failure to curb emissions would directly affect climate change, and thus hurt the U.S., too.

The Tragedy of the Commons

Cooperation problems have always been present in human societies. One example is that of the tragedy of the commons. In English villages before the 16th century, parcels of land, called the commons, that belonged to a collection of individual farmers were accessible to the whole village. While all villagers could let their sheep graze on the commons, individual villagers had an incentive to let their sheep overgraze there without feeling responsible for the potential damage to the land. No matter what the other villagers decided to do, an individual's maximum self-interest lay in letting his sheep graze as much as possible on the commons so that each villager would have to secure less feed for his sheep. Of course, this self-interest led to serious overgrazing.

In the 16th and 17th centuries, there was an "enclosure" movement launched by the British landed aristocracy to keep the commons away from those who did not have property claims, usually the poor peasants. This movement had major social effects on the English countryside and resulted in large numbers of peasants

becoming severely impoverished and leaving their villages for the cities.

We do not know exactly how much overgrazing there was in England at the time, but the tragedy of the commons is an illustration of a more general phenomenon that has occurred over and over in history, the overexploitation of arable land and the depletion of natural resources. The "fertile crescent" in the Middle East, around the Nile, Jordan, Tigris, and Euphrates rivers, was one of the most arable and productive areas of the ancient world. Overexploitation has eroded the quality of the soil and led to desertification. Similar problems have occurred in other regions due to deforestation; the population of Easter Island nearly became extinct following its complete exploitation of the island's native tree species.⁸ Deforestation of the Amazon rain forest in South America, the major producer of oxygen on the planet, is occurring at a phenomenal speed; the annual loss of rain forest in the Amazon region is roughly 10,000 square miles, nearly the size of Massachusetts.

The collective-action problem. Another important example of the prisoner's dilemma is the **collective-action problem**, when people fail to undertake a collective action even though it is in their joint interest to do so. Imagine an Indian municipality's decision to authorize the construction of a chemical factory that will pollute the town's air and water, and suppose that the residents are opposed to the decision. To defeat the project, the population needs to be organized, but doing so is costly. Assume that if at least half the population shows up at a demonstration to protest the decision, the project will be abandoned. If, instead, less than half of the population shows up, the project will go forward. Demonstrating is costly and inconvenient, however, and much of the population would rather spend their free time elsewhere. Just as in the prisoner's dilemma, if very few demonstrate, participation will not be effective and it will not be optimal for anyone to demonstrate. However, if at least half of the population goes to the demonstration, any resident can **free ride** and benefit from the

⁸See, for example, James A. Brander and M. Scott Taylor, "The Simple Economics of Easter Island: A Ricardo-Malthus Model of Renewable Resource Use," *The American Economic Review* 88, no. 1 (March 1998): 119–138.

actions of the others while not incurring any of the costs of taking action. Thus, it is a dominant strategy for an individual not to demonstrate and the result is that no one demonstrates and the project is implemented, an outcome that is suboptimal for the residents. Everyone wants the demonstration to happen, but no one wants to participate individually, leading to collective-action failure.

Collective-action failures are fundamental to understanding human societies. They help to explain why inefficient institutions persist in developing countries. We frequently read in the press of African dictators who loot their countries' resources and put the proceeds, sometimes billions of dollars, in Swiss banks. Why do such situations continually occur? It is precisely because the collective-action problem prevents people from overthrowing corrupt leaders or dictatorial regimes.

Institutional solutions to the cooperation problem. Institutions are necessary to solve cooperation problems. Historically, labor unions in many countries have helped workers organize collective actions in order to obtain better working conditions and higher wages. Professional organizations such as associations of farmers, fishermen, construction workers, and doctors defend the interests of their professions in both the legislative process and the media. Political parties serve as organizations that promote competing political platforms via the elections of party members and the passage of legislative bills.

International institutions can also solve the cooperation problem. The United Nations (UN) was founded after World War II to prevent similar global conflicts from ever occurring again. Unfortunately, the UN was not given much power by the sovereign countries that make up its membership. This has prevented the organization from becoming an effective and powerful international institution able to intervene effectively in various world conflicts. The European Union (EU) is a more successful institution than the UN. European nations have been at war with each other for most of the last 1,000 years, but the creation of the EU was seen by Europeans as an institutional solution to cooperation problems in the region. Institutional solutions to the environmental problems of the planet will require new forms of international cooperation in which world leaders meet and jointly agree on solutions, and then commit to their implementation.

Economists do not completely understand how the myriad associations and organizations that we observe in the world are able to overcome collective-action problems. People are not only organized into political parties, trade unions, and professional associations at the national level, but they also organize to cooperate at the international level. There are many small-scale examples of spontaneous collective action as well, such as villagers organizing to rebuild their homes after a fire or a flood. According to standard game theory, these associations and organizations should not emerge spontaneously because of the collective-action problem. It takes very dedicated individuals to incur the heavy cost of spending time and money to set up these cooperative efforts. The members of nongovernmental organizations (NGOs) often feel passionately about the causes for which they fight, whether it is the environment, the eradication of poverty, or advancements in world health, in contrast to people who take a more passive approach to these problems and do not choose to spend time, energy, or money to organize and participate in new problem-solving associations.

Elinor Ostrom and Institutional Solutions to the Preservation of Common Resources

In 2009, Elinor Ostrom was the first woman to receive the Nobel Prize in Economics, which was jointly awarded to Oliver Williamson, one of the most famous researchers working on institutions. Ostrom's best-known work focuses on the emergence of institutions to solve tragedy-of-the-commons situations, in particular the management of common resources. Examples include water supplies, fish stocks, forests, and grazing lands. Economists automatically tend to think of two basic solutions to manage common resources: privatize them by selling rights to access them or let a central government regulate access to them. Ostrom found that in many of these situations, structures emerge from local governments that prevent overexploitation of

common resources. Local communities establish rules to regulate access and put in place conflict-resolution mechanisms to enforce the rules and monitor resource use. For example, farmers would take turns to graze cows or utilize scarce water. People in the community would create boundaries where wood cutting was allowed, etc. These local structures also enforce sanctions for violation of the rules. The governance structures generally work well when they are based on democratic decision-making processes and are most effective in smaller communities where people know and trust each other, and interact regularly. It is also easier in smaller communities to enforce collective sanctions on individuals who break the rules.

Democracy and the collective-action problem. Political institutions can either help to overcome collective action problems or can exacerbate them. In dictatorships, the leaders usually ban political parties and associations; as a result, the opposition must organize secretly. Punishment for establishing or participating in opposition groups may mean death or lengthy jail time.

In democracies, freedom of association is a fundamental right. Professional associations and NGOs even receive assistance from the government in the form of grants and subsidies, even though they are associated with the government. In other words, democracy allows and even encourages the establishment of associations to overcome collective-action problems. Moreover, elections, the primary institution of democracy, provide a way to overcome one of the most fundamental collective-action problems faced by societies, replacing corrupt or inefficient leaders with better leaders. In a democracy, people come to the polls and can vote out incumbent leaders who no longer represent voters' interests or who have abused their powers.

Informal solutions to the cooperation problem. Most solutions to the cooperation problem tend to be formalized in some way. For example, trade unions, professional associations, and NGOs have their own rules and organizational structures that ensure the groups' proper functioning. This does not mean that there are no informal solutions to the cooperation problem. For example, spontaneous riots can break out in a developing country that can potentially lead to political change. Mine workers might engage in a spontaneous strike without consulting trade-union leaders. In 1989, Protestant activists started small

demonstrations against the East German Communist regime in Leipzig. Within a few weeks, these demonstrations were joined by tens of thousands of East German citizens who had never before demonstrated. Two years previously, a similar uprising occurred, protesting the dictatorship in South Korea. The country had been living under various forms of dictatorship since the end of the Korean War. Student organizations began actively fighting for democracy and, in 1987, nationwide protests erupted, eventually leading to the downfall of the dictatorship.

Large-scale riots are usually spontaneous and the result of some dramatic event such as a hike in food prices or the political assassination of an opposition leader. These forms of collective action are usually unpredictable and not well understood by social scientists. Because they are spontaneous and not backed by a formal organization, these collective actions are also fragile and their momentum is often difficult to sustain. Participants may demobilize very rapidly and the mass movement may die out without having achieved any precise goal. Sometimes, these mass movements are however more powerful and sustained than expected and lead to revolutionary change, like in the case of the Arab Spring.

The Coordination Problem

Game theorists define a coordination problem as a situation in which agents can all coordinate on a specific action and the action an individual chooses depends on what others do. There are **multiple equilibria** possible in coordination problems, different possible sets of actions on which individuals coordinate. Imagine a situation in which there are no motor vehicle statutes and individuals must decide on their own whether to drive on the right or on the left. All individuals would quickly unite to drive on one side or the other. However, we do not know on which side they would choose to drive. In terms of game theory, these choices are both Nash equilibria. If an individual sees everyone else driving on the right, he has an incentive to drive on the right as well, but if he sees all others driving on the left, he will have an equally strong incentive to drive on the left. Once everyone drives either on the left or on the right, no one will have an incentive to deviate because that would be dangerous. The only thing we can predict is that people will decide to adopt a single rule addressing on which side to drive.

The difference between coordination and cooperation problems. The difference between these problems lies in the fact that in a coordination problem, what is optimal for an individual depends on what the others do. This is why there are multiple equilibria. There is no longer a dominant strategy as in the prisoner's dilemma where there is only one possible equilibrium, the inefficient one. In the case of the driving rule, none of the two equilibria (everybody driving to the left or everybody driving to the right) seems better or worse than the other, assuming that steering wheel can be adjusted to the prevailing equilibrium. There are many coordination problems in which all equilibria give the same payoffs to agents. In some countries, when people decide to meet at the train station, the meeting place is below the main clock. In others, it is at the main exit, by the cab station or on the platform.

There are also coordination games with equilibria that are not all equally efficient. Imagine the very relevant situation in developing countries where people can either decide to be law-abiding citizens or to engage in illegal activities. They would choose to be law-abiding citizens if there were a significant chance that they would be caught in the commission of a crime, but would engage in illegal activities if there were a low chance of being apprehended. The police have limited resources to fight crime, so the probability of being caught depends on the number of people engaged in illegal activities. If few people break the law, then the probability of being caught is high and it is preferable to be honest, but if many break the law, the probability is lower and there are more incentives to engage in illegal activities. We have two equilibria: one in which people are honest under the rule of law and one in which people engage in illegal activities with no adequate law enforcement, and thus no rule of law. If the rule of law is a condition for productive investment to take place, which is crucial for growth, then the rule of law equilibrium will be better for citizens in the long run. Here again, an individual's choice depends on the choice of others.

Understanding coordination problems is important in the context of development for a variety of reasons. First, the social norms that have emerged in various countries are quite diverse. Norms of politeness vary greatly among countries and a lack of knowledge of these norms can easily lead to social misunderstandings. In some Asian countries, for example, entering a house or even a restaurant without taking off your shoes is extremely impolite. Norms of conversation in parts of East Africa require repeating what the previous speaker said before expressing one's own opinion.

Second, we can observe in many countries coordination on a suboptimal equilibrium with a low level of law abidance, high levels of tax evasion, a large informal sector, and a high level of acceptance for corruption. Reforms that help a country shift from one equilibrium to the other are very difficult to design and implement because they require a collective effort that, as seen above, can be difficult to achieve. Diagnosing the problem is often easier than finding an effective and practical solution.

Coordination and "big push" strategies. We can look at the "big push" strategies of structural change discussed in Chapter 5 from the perspective of coordination problems. If there is little infrastructure (ports, roads, or electricity), private investors will be reluctant to invest in a country because their investment will not be very productive without infrastructure. A government, in turn, will not want to spend money on infrastructure if it thinks that there will be little private investment. This results in a coordination equilibrium, where there is minimal infrastructure and minimal private investment.

We can interpret the "big push" idea as an attempt to move from that low investment equilibrium to one in which there are high levels of both infrastructure and private investment. If there is a well-established infrastructure, private firms will be more willing to invest in the country and the government will be more willing to invest in the infrastructure if it thinks there will be significant private investment. The "big push" strategy consists of providing aid for infrastructure investment while simultaneously developing policies to encourage private investment.

The Stag Hunt Game

The stag hunt game is a coordination problem that was first formulated by the 18th-century French philosopher, Jean Jacques Rousseau (1712–1778). Economists and social scientists often use it as the prototype for all the coordination problems faced by societies and other groups in the real world. The following table illustrates the stag hunt game.

	A: Stag	A: Hare
B: Stag	4, 4	0, 3
B: Hare	3, 0	3, 3

Two hunters, A and B, are stalking game in a forest. Each can either choose to hunt a hare or a stag. A stag is more valuable than a hare but will only be caught if A and B hunt together. In

this case, they both get a payoff of 4. If a player chooses to hunt a hare, he will get a payoff of 3. However, a player who chooses to hunt a stag while the other hunts a hare will not be successful and will have a payoff of 0. This table shows the benefits of hunting together because the payoff for hunting a stag together is higher than hunting for a hare individually. This is a coordination game, since it has multiple equilibria (both hunt a stag and both hunt a hare) and there is no dominant strategy: if one player hunts a stag, then the other will choose to do the same, but if one player hunts a hare, then the other player will also hunt a hare.

We can easily use the structure of the stag hunt game to examine issues that are relevant to developing countries, such as high or low levels of coordination in terms of law abidance or tax evasion.

Institutional solutions to coordination problems. In developing countries, formal laws help individuals and groups select one coordination equilibrium over another. This is the case for the traffic law example. The regulation of standards is another important solution to the coordination problem; railroads, for example, have different track widths in different countries, but most nations have implemented the international standard track width of 56.5 inches (1,435 mm). India and Pakistan instead inherited a broader gauge of 66 inches (1,676 mm) from the British colonial period, while Russia and the other nations of the former Soviet Union, as well as Mongolia, adopted a gauge of 59.84 inches (1,520 mm). Differences in gauge across countries tend to persist because of the costs of standardization. This requires passengers to change trains at international borders or await an axle adjustment, both of which are costly.

Informal solutions to coordination problems. Conventions, a type of social norm, are enforced informally by peer pressure and experience, and do not require legal enforcement. For example, when a telephone conversation is interrupted, the caller is usually the one to recall. Otherwise, each caller would be calling or no one at all would be calling back.

In 2005, Thomas Schelling received the Nobel Prize in Economics for his studies of coordination problems. He put forward the concept of **focal point** to understand how people can coordinate their actions. A focal point is a solution to a coordination problem that most people will find without having to communicate. For example, if two people decide to meet in a big city like Rio de Janeiro, but forget to specify where and do not have the opportunity to call

each other again before their appointment, it is likely that both will go to the Central Station. This is an example of a focal point.

Conventions are usually well understood within a country or a particular culture, but not always across countries. In many cultures, a nod of the head up and down means "yes," while a movement of the head back and forth from left to right means "no." When greeting someone, Americans shake hands, but the Japanese bow. The exchange of business cards is fundamental to business etiquette in most of Asia, but not in many other countries. In the Middle East, showing the sole of your shoe to somebody else is considered an insult. Similar body movements or actions can have different interpretations in different societies. There are hundreds, probably thousands of social cues that differ across cultures and can make cross-cultural communication difficult, even when the language barrier is not significant. It is always respectful in a foreign country to show the locals that you have some knowledge of their local customs and conventions.

Many social norms are not necessarily fundamental for economic development. Some norms may, however, play a very important role: norms of honesty and trust, norms of conflict resolution (peaceful or violent), norms of equality between men and women, norms related to individual savings behavior, or norms related to effort. We will discuss many of these issues in Chapter 11 when we discuss the effect of culture on development.

The Persistence of Inefficient Institutions

We have seen how institutions help solve economic problems that stem from lack of information, commitment, and cooperation or coordination problems. While developed countries have institutions to solve these problems, developing countries are often not as fortunate. The lack of formal institutions impedes economic growth by raising the costs of transactions and slowing the flow of capital into a developing country. If improving institutions is crucial for successful development, we should ask why it is not happening. Reforming institutions is not as simple as signing a contract between two individuals. Inefficient institutions can even persist for a very long time and for a variety of complex reasons.

The Functionalist Fallacy

In this chapter, we have seen what institutions do, but not why institutions are in place. Understanding the functions that institutions perform is one thing; understanding why they emerge is another.

Institutions, just like technical innovations, are often the unintended consequences of actions designed to solve other problems. We know what microwave ovens do; they have become indispensable household appliances to warm or defrost food quickly. The microwave oven can warm a meal much faster than a traditional oven but that was not the purpose its inventor had in mind. It was an accidental byproduct of research to make better radar systems. Dr. Percy Spencer, an engineer working for the Raytheon Corporation, was testing a vacuum tube called a magnetron while researching improvements to conventional radar when he noticed that a candy bar in his pocket had melted. He put an egg and some corn kernels near the tube and found that those foods

cooked quickly when exposed to microwaves. Thus, there can be a significant difference between the original purpose of an invention and its ultimate practical function, which in the case of the microwave, was quite accidental. This is also the case for many inventions such as silicon, saccharin, and the cardiac pacemaker among many others.

A similar process has occurred for many institutions. Settlers who came to North America or Australia did not have in mind the goal of creating institutions for successful economic development. They were primarily focused on preventing the creation of a monarchy such as the one they had left in England. Freedom from tyranny was their objective. They had no idea that the institutions they created would lead to phenomenal economic success.

This discussion illustrates the idea that function does not constitute an explanation for existence. There is the danger of the **functionalist fallacy**, which mistakes an object's function for the cause of its existence. Why is this distinction important? If we did not pay attention to this problem, we would tend to adhere to the view that institutions are always efficient. Indeed, if institutions solve certain problems, we would tend to believe that once a problem emerges, an institution will automatically appear to solve it. If this were the case, institutions would always be efficient. In fact, many institutions are typically inefficient, and inefficient institutions can persist for a very long time. Bringing about change to institutions requires overcoming the collective-action problem. This is very difficult to do in developing countries, especially if democratic institutions are absent or imperfect.

We will take a closer look at inefficient institutions throughout this book. Corruption is rife in many developing economies, with inefficient courts, dictatorial regimes, little protection for property rights, and a lack of formal titling systems. Inefficient formal institutions are very hard to overcome, especially when they aim to prevent agents from organizing to change them. Until considerable individual courage and group mobilization can overcome inefficient institutions, they may persist for a very long time.

Leaders in Institutionalism

The expansion of institutionalism that has taken place in economic theory is a revival of sorts. Economists developed contract theory and principal agent theory to analyze problems with individual transactions between agents who are in a contractual relationship. Game theory provides the tools to consider such problems analytically. Nevertheless, before game theory was developed, many economists already took institutions very seriously even though they were not part of mainstream theory. Throughout the 20th

century, quite a few famous economists tried to understand the role of institutions and how they develop. We mention some of these prominent economists below.

Thorstein Veblen (1857–1929), in a famous book called *The Theory of the Leisure Class* (1899), coined the concept of **conspicuous consumption**, explaining that consumption among the rich was dictated more by a desire to show one's wealth and status rather than by the useful characteristics of the good consumed. He also believed that

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much of human behavior was influenced by cultural and social norms.

John R. Commons (1862–1945) believed that transactions were the basis of any economic activity. He was interested in the legal and institutional aspects of transactions, and defined an institution as a collective action. He was one of the first economists to use the concepts of bounded rationality, incomplete contracts, and property rights, which became very prominent in institutionalist thinking.

Ronald Coase (1910–) is a British economist who spent his career at the University of Chicago. His research centered on the notion of transaction costs, which he considered to be the reason why firms exist and why we observe many transactions taking place within a firm rather than across legal entities. His 1937 article, “The Nature of the Firm,” is one of the most influential writings in economics. His 1960 article, “The Problem of Social Cost,” in which he developed the famous Coase theorem, states that private contracts can efficiently deal with externalities as long as there are well-defined property rights in place. He strongly encouraged the development of research at the intersection between Law and

Economics and received the Nobel Prize in Economics in 1991.

Oliver Williamson (1932–) developed transaction-cost economics and his ideas gained wide acceptance among mainstream economists. He also developed the idea of the hold-up problem and of asset-specific investments, as well as many other related concepts. He received the Nobel Prize in Economics in 2009.

Douglass North (1920–) is an economic historian who emphasized the role of institutions in securing property rights and in explaining economic success over the long term. He received the Nobel Prize in Economics in 1993.

Mancur Olson (1932–1998) was the first economist to put forward the collective-action problem and to explain that individuals could free ride on the efforts of others. He developed the idea that collective action is like a public good and will be underprovided by only private efforts. He predicted that large groups with a minimal stake in public action will tend to suffer more from the collective-action problem than will small groups with a very large stake in public action. His work was seminal in understanding why inefficient institutions may persist.

Chapter Summary

What Are Institutions?

We have introduced the concept of institutions and shown that institutions matter for economic development. We have defined institutions as constraints on human behavior imposed by societal rules. Formal institutions, such as legal and political institutions, are written rules with codified statutes of enforcement verified by courts. Informal institutions are social norms or patterns of behavior dictated by cultural or religious beliefs. Informal rules are enforced in an informal way by peer pressure, a sense of moral obligation, or by informal enforcement agencies.

What Do Institutions Do?

Institutions play a role in solving problems inherent to economic and social transactions: informational problems, commitment problems including the hold-up problem, cooperation, and coordination problems.

Informational problems arise most often when one of the parties to a transaction has information about the object of the transaction that the other party does not have. Informational problems are an impediment to efficient market transactions. Institutional solutions to informational problems are, for example, disclosure rules, warranties, return

policies, or the emergence of informational intermediaries such as rating agencies and chain stores. Informal institutional solutions to informational problems involve the reputation of economic agents and the circulation of information about their reputations.

Commitment problems arise when a party in a transaction has an incentive to renege on a promise made to the other party. Formal institutions solve the commitment problem by enforcing legal contracts. Informal enforcement of commitments is more difficult and sometimes requires the use of criminal organizations.

Cooperation problems arise because agents must incur individual costs when participating in any form of joint action, but can benefit from the effort of others even when they do not participate in the action. As a result, collective action occurs less often than is necessary. Formal institutional solutions to collective action include political and civil associations. Democracy is an important solution to

collective action by means of elections, which allow voters to make collective choices for leaders and policies. Informal solutions include riots, spontaneous strikes, and other actions that occur in special circumstances.

Coordination problems arise when the actions of some agents affect the actions of others and when agents coordinate on positive behaviors (behaving honestly, trusting others, paying taxes) or negative behaviors (behaving dishonestly, not trusting others, evading taxes). Formal solutions to coordination problems involve laws and standards; informal solutions involve social norms and conventions.

The Persistence of Inefficient Institutions

Inefficient institutions have negative long-term effects on the economy. They may, however, persist for a very long time precisely because reforming institutions requires overcoming cooperation and collective-action problems.

Key Terms

adverse selection
asset specificity
asymmetric information
civil-law systems
collective-action problem
commitment problem
common-law system
conspicuous consumption
cooperation problem
Copenhagen Accord
credibility
credit rationing
creditworthiness
culture
disclosure rules
dominant strategy
expected payoff
focal point
formal institutions
free ride
functionalist fallacy
game theory
hold-up problem
informal institutions

informational intermediaries
institution
Kyoto Protocol
moral hazard
multiple equilibria
Nash equilibrium
prisoner's dilemma
rating agencies
regulation of access or entry
relational contracting
relationship specific
return policies
risk averse
risk neutral
sequential transactions
signaling
social norms (conventions)
sunk cost
time inconsistency
transaction costs
values
vertical integration
warranty

Review Questions

1. Give an example (different from those discussed in the chapter) of an economic transaction that is more difficult to complete in developing countries than it would be in developed countries.
2. Many solutions to informational problems emerge spontaneously as contractual innovations. Is there a role for the formal institution of law with respect to these contractual innovations, and if so, why?
3. Reputation is important for contract enforcement when formal institutions are weak. Explain. What conditions in the economic environment make reputation work more or less efficiently? Why?
4. Give examples of formal and informal institutions as solutions for the cooperation problem that we have not discussed in the chapter. Explain your choices.
5. Why is the coordination problem important in the context of development?
6. Write down the problem of cooperation for two countries, A and B, in the fight against climate change and use a payoff matrix as in the case of the prisoner's dilemma. Payoffs are $y_A(x_A, x_B)$ and $y_B(x_A, x_B)$, where x_A and x_B can take two values: 1) if measures are taken in the country against climate change and 2) if no measures are taken. What are the conditions of the payoffs in each cell for each player so that there is a unique Nash equilibrium in which no country takes any action against climate change?
7. Assume a coordination problem related to honest business practices, based on the following payoff matrix:

	B is honest.	B is dishonest.
A is honest.	a, a	d, b
A is dishonest.	b, d	c, c
8. What are the conditions on the payoff matrix so that both A and B being honest and dishonest are both equilibria? How would you have to change the payoffs so that both A and B being dishonest is the only equilibrium? What change would that represent in the problem analyzed?
8. Assume that an Indian firm is contacted by a multinational car company to produce brake systems. This requires an initial investment of I from the Indian subcontractor and a variable cost c per braking system. The car company plans to buy n brake systems from the subcontractor. What is the price p at which the Indian subcontractor would break even? At what price p' can the car company still convince the subcontractor to produce the systems once it has made the investment?
9. Look up the following article by Simon Brinsmead: "Oil Concession Contracts and the Problem of Hold-Up," downloadable at http://www.dundee.ac.uk/cepmlp/journal/html/Vol17/Vol17_11.pdf. How do the ideas in this article relate to this chapter's discussion? Assume you were advising the government of a developing country with oil reserves. What kind of contract would you recommend to the government? Explain.
10. Download the governance indicators from the World Bank (click on Download full dataset) in either excel or stata format: <http://info.worldbank.org/governance/wgi/index.asp>. Look at the Rule of Law index ranking of countries in 2010 and in 1996. What observations do you make when comparing these two rankings?